

Math 54 ReviewName: _____ ID: _____

Part I: Linear Systems (Chapter 1)**1. Systems and Row Reduction (1.1–1.2)****1.1** Solve the system

$$\begin{cases} x + 2y - z = 1, \\ 2x + 5y - 3z = 4, \\ x + 3y - 2z = 2. \end{cases}$$

(a) Write the augmented matrix. (b) Row reduce. (c) Give the solution set (or state none).

1.2 Determine whether the system is consistent. If consistent, describe *all* solutions.

$$\begin{cases} x - y + z = 0, \\ 2x - 2y + 2z = 3. \end{cases}$$

1.3 Row reduce the augmented matrix to RREF and solve:

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 2 & 4 & 7 & 3 \\ 1 & 2 & 4 & 2 \end{array} \right].$$

Identify pivot columns and free variables.

1.4 For which values of k is the system consistent? For consistent k , describe the solution set.

$$\begin{cases} x + y + z = 1, \\ 2x + 2y + 2z = k. \end{cases}$$

2. Vector Equations and $Ax = b$ (1.3–1.5)**1.5** Decide whether

$$b = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} \quad \text{is in} \quad \text{span} \left\{ v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

If yes, find scalars c_1, c_2 such that $c_1v_1 + c_2v_2 = b$.

1.6 Let

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 3 \end{bmatrix}.$$

- (a) Compute Ax for $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ and rewrite Ax as a linear combination of the columns of A .
- (b) Describe all $b \in \mathbb{R}^2$ for which $Ax = b$ is consistent.
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1.7 Solve the homogeneous system and give the solution in parametric vector form:

$$\begin{cases} x + 2y - z = 0, \\ 2x + 4y - 2z = 0. \end{cases}$$

1.8 (Structure of solutions) Consider

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}.$$

- (a) Find one particular solution x_p to $Ax = b$.
- (b) Find the general solution to $Ax = b$ in the form $x = x_p + x_h$, where $x_h \in \text{Null}(A)$.
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3. Linear Independence and Linear Transformations (1.7–1.9)

1.9 Determine whether the vectors are linearly independent:

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

If dependent, exhibit a nontrivial linear relation.

1.10 Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by $T(x, y) = (2x - y, x + 3y)$.

- (a) Find the standard matrix A such that $T(\mathbf{x}) = A\mathbf{x}$.
- (b) Compute $T(1, 2)$.
- (c) Determine whether T is one-to-one and/or onto.
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1.11 Find the standard matrix of $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$T(x, y) = (x + 2y, 3x - y).$$

Then compute $T(2, -1)$.

Part II: Matrix Algebra (Chapter 2: 2.1–2.3)

2.1 Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}.$$

Compute $A + B$, $3A$, AB , and BA . Are AB and BA equal?

2.2 Compute A^2 and A^3 for

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

Conjecture a pattern for A^{2n} and A^{2n+1} .

2.3 Find A^{-1} (if it exists) using row reduction:

$$A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}.$$

Then solve $Ax = b$ for $b = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$ using A^{-1} .

2.4 (Concept) True or False. Justify briefly.

- (a) If the columns of A are linearly independent, then A is invertible.
 - (b) If A is invertible, then $Ax = b$ has at most one solution for every b .
 - (c) If $Ax = 0$ has only the trivial solution, then A is invertible.
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Part III: Determinants (Chapter 3: 3.1–3.3)

3.1 Compute

$$\det \begin{bmatrix} 1 & 2 & 0 \\ 3 & 4 & 5 \\ 0 & 1 & 2 \end{bmatrix}$$

using cofactor expansion (choose a convenient row/column).

3.2 Without fully expanding, compute $\det(A)$ for

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & 4 \\ 0 & 0 & 5 \end{bmatrix}.$$

Explain why your method works.

3.3 Use determinant properties to compute $\det(B)$ given $\det(A) = 6$:

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix} A.$$

3.4 Use Cramer's rule to solve

$$\begin{cases} 2x + y = 5, \\ 3x - y = 4. \end{cases}$$

3.5 (Determinant and invertibility) Consider

$$C = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{bmatrix}.$$

Compute $\det(C)$ quickly (hint: look for row dependence). Is C invertible?

Part IV: Vector Spaces (Chapter 4: 4.1–4.2)

1. Vector Spaces and Subspaces (4.1)

4.1 Is the set

$$S = \{(x, y) \in \mathbb{R}^2 : x + y = 1\}$$

a subspace of $\mathbb{R}^2$? If not, identify which subspace property fails.

4.2 Let

$$W = \{(x, y, z) \in \mathbb{R}^3 : x - 2y + z = 0\}.$$

- (a) Prove W is a subspace of $\mathbb{R}^3$ using the subspace test.
 - (b) Find a parametric vector description of W .
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2. Null Space and Column Space (4.2)**4.3** For

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix},$$

find $\text{Null}(A)$ and $\text{Col}(A)$, and compute $\text{rank}(A)$.

4.4 Let

$$M = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

- (a) Find a basis for $\text{Col}(M)$.
 - (b) Find a basis for $\text{Null}(M)$.
 - (c) Verify $\dim(\text{Col}(M)) + \dim(\text{Null}(M)) = 3$.
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